Awesome Languages and Cloud Foundry

Awesome Languages

Clojure

Golang

Elixir

Erlang

Community

The way we deploy apps is changing

From hardware to APIs and instances

Containers running on those instances

Applications are more ~~heterogenous~~ cobbled together

Yay, new abstractions!

Cloud Foundry

Deployment

```
$ cf push my-app
```

Distributes my app across a large number of instances

Abstracts service dependencies

Monitors and respawns instances

Avoid a bunch of deployment concerns

Awesome Languages & Cloud Foundry

Buildpack support

Transparent service binding

Scale applications

Extend the platform in the language of our choice

Buildpack Support

Sinatra

```
get '/cats/:name' do
  json name: params['name'],
       breed: :tabby
end
```

Sinatra Push

```
$ cf push cats-sinatra
-----> Downloaded app package (4.0K)
-----> Using Ruby version:
ruby-2.0.0
-----> Installing dependencies using
Bundler version 1.3.2
       Running: bundle install --
without development:test --path
vendor/bundle --binstubs vendor/
bundle/bin --deployment
```

Sinatra Push

```
-----> Uploading droplet (52M)
Registering instance

0 of 1 instances running (1 down)
0 of 1 instances running (1 starting)
Started
```

Sinatra Curl

```
$ curl -D- 'http://cats-
sinatra.example.com/cats/stevens'
HTTP/1.1 200 OK
Content-Type: application/
json;charset=utf-8
Date: Thu, 5 Dec 2013 14:30:50 GMT
Server: thin 1.6.1 codename Death Proof
X-Content-Type-Options: nosniff
Content-Length: 34
Connection: keep-alive

{"name":"stevens","breed":"tabby"}
```

Sinatra Instances

```
$ cf apps
Getting apps in org cloudcredo /
space codemesh as admin...
OK

name:       cats-sinatra
state:      started
instances: 1/1
memory:     128M
disk:       1G
urls:       cats-sinatra.example.com
```

Sinatra Instances

```
$ cf scale cats-sinatra -i 10
Scaling app cats-sinatra in org
cloudcredo / space codemesh as
admin...
OK
```

Sinatra Instances

```
$ cf app cats-sinatra

state: started
instances: 10/10
usage: 128M x 10 instances
urls: cats-sinatra.example.com

     status    cpu    memory          disk
#0   running   0.0%   20.3M of 128M   121.4M of 1G
#1   running   0.0%   20.3M of 128M   121.4M of 1G
#2   running   0.0%   20.3M of 128M   121.4M of 1G
#3   running   0.0%   20.3M of 128M   121.4M of 1G
#4   running   0.0%   20.3M of 128M   121.4M of 1G
#5   running   0.0%   20.3M of 128M   121.4M of 1G
…
```

CF_TRACE

```
$ CF_TRACE=true cf app cats-sinatra

{
  "detected_buildpack":"Ruby/Rack",
  "memory":128,
  "instances":1,
  "disk_quota":1024,
  "state":"STARTED"
}
...
```

Liberator

```
(defresource cat [name]
  :available-media-types
    ["application/json"]
  :handle-ok
    (fn [_]
      {:name name :breed :tabby}))

(defroutes app
  (ANY "/cats/:name" [name]
    (cat name)))
```

Liberator Push

```
$ cf push cats-liberator -b https://
github.com/heroku/heroku-buildpack-
clojure.git
-----> Installing OpenJDK 1.6...done
-----> Installing Leiningen
       Downloading: leiningen-2.3.4-
standalone.jar
       Writing: lein script
-----> Building with Leiningen
       Retrieving org/clojure/clojure/
1.5.1/clojure-1.5.1.pom from
…
```

Liberator Push

```
-----> Uploading droplet (66M)
Registering instance

0 of 1 instances running (1 down)
0 of 1 instances running (1 down)
0 of 1 instances running (1 starting)
0 of 1 instances running (1 starting)
0 of 1 instances running (1 starting)
0 of 1 instances running (1 starting)
Started
```

Liberator Curl

```
$ curl -D- 'http://cats-
liberator.example.com/cats/stevens'
HTTP/1.1 200 OK
Content-Type: application/
json;charset=UTF-8
Date: Thu, 5 Dec 2013 14:30:50 GMT
Server: Jetty(7.6.1.v20120215)
Vary: Accept
Content-Length: 34
Connection: keep-alive

{"name":"stevens","breed":"tabby"}
```

CF_TRACE

```
$ CF_TRACE=true cf app cats-liberator

{
  "buildpack": "https://github.com/
heroku/heroku-buildpack-clojure.git",
  "detected_buildpack":null,
  "memory":128,
  "instances":1,
  "disk_quota":1024,
  "state":"STARTED"
}
…
```

bin/detect

bin/compile

bin/release

Martini

```
m.Get("/cats/:name", func(
  w http.ResponseWriter,
  params martini.Params) string {
    w.Header().Set("Content-Type",
                   "application/json")
    var cat = struct {
      Name  string `json:"name"`
      Breed string `json:"breed"`
    }{params["name"], "tabby"}
    b, _ := json.Marshal(cat)
    return string(b)
  })
```

Martini Push

```
$ cf push cats-martini -b https://
github.com/hmalphettes/heroku-
buildpack-go.git
-----> Downloaded app package (1.8M)
Initialized empty Git repository in /
tmp/buildpacks/heroku-buildpack-
go.git/.git/
-----> Installing Go 1.1.1... done
-----> Running: go get -tags
heroku ./...
-----> Uploading droplet (3.1M)
```

Martini Curl

```
$ curl -D- 'http://cats-martini.example.com/cats/shaft'
HTTP/1.1 200 OK
Content-Type: application/json
Date: Thu, 5 Dec 2013 14:30:50 GMT
Content-Length: 32
Connection: keep-alive

{"name":"shaft","breed":"tabby"}
```

Dynamo

```
get "/cats/:name" do
  conn = conn.resp_content_type(
    "application/json")
  conn.resp 200,
    JSON.generate [name: name,
                   breed: "tabby"]
end
```

Lies, Damned Lies

```
$ ab -c 10 -n 1000 $URL
```

	50%	90%	95%	100%
sinatra	13	33	37	69
liberator	8	35	46	83
martini	9	38	44	58

Derek Collison, Sept 2012

“Prediction: Go will become the dominant language for systems work in IaaS, Orchestration, and PaaS in 24 months. #golang"

Docker

etcd

Cloud Foundry

Serf

Packer

Cloud Foundry Implementation

Loosely coupled components talking over a message bus (NATS)

Historically Ruby with Event Machine for Concurrency

Number of core components now ported to Golang

Golang Advantages over Ruby

Better support for concurrency (goroutines)

Deployment

Performance

cloudfoundry/gorouter

Responsible for routing incoming requests to app instances

V1: Nginx with embedded Lua talking to Ruby

V2: Standalone Go Binary

cloudfoundry/cli

End-user client utility

Ruby distribution headaches

Much snappier

apcera/gnatsd

NATS daemon in Go

Added support for clustering

Released in October

Performance

cloudfoundry/dea_ng

vito/garden

Container Isolation

Golang implementation of Warden

Still under development

ActiveState Stackato uses Docker

cloudfoundry/hm9000

Compares desired and actual state

Originally Ruby and Eventmachine

Previously single-instance only

Nearing readiness for production

Service Brokers

Responsible for provisioning new service instances

MySQL, RabbitMQ ...

actual service

⇧

<u>service broker</u>

⇧

cloud controller

⇧

punter

Typically implemented by extending a Ruby Base Class

Old API featured bi-directional communication with the Cloud Controller

V2 API makes communication one-way

Series of REST calls to your broker

Write your broker in anything you like

Broker-ception - host your broker on CF itself

Wrapping it up

☺ Make deployment easy to help community adoption

☺ Cloud Foundry is fun to hack on

Thanks. We're Hiring!

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